

CONTACT INFORMATION	Team EPILOGY	Google Scholar: Tat-Thang Vo
	Institut Mondor de Recherche Biomedicale INSERM U955. University Paris Est Créteil.	Linkedin: Tat Thang Vo ✉ E-mail: tat-thang.vo@u-pec.fr
ACADEMIC POSITIONS	Chair of junior professor in biostatistics - University Paris Est Créteil (France) - Mondor Institut of Biomedical Research, INSERM U955 - Research Group EpiDerme - Epidemiology in Dermatology and Evaluation of Therapeutics.	9/2023–now
EDUCATION	Ghent University (Belgium) and Université Paris Cité (France) - Ph.D. in Biostatistics. - Topic: Causal inference in meta-analysis and mediation analysis. - Advisors: Pr. Stijn Vansteelandt and Pr. Raphael Porcher.	2016–2020
	University Paris Cité (France) - M.Sc., Comparative Effectiveness Research. First-class honors. - ConEd., Clinical Pharmacy. First-class honors.	2014–2016
	The Open University (UK) B.Sc.(Hons), Mathematics. First-class honors.	2017–2020
	Hanoi University of Pharmacy (Vietnam) Pharm.D., Clinical Pharmacy. First-class honors. Valedictorian.	2009–2014
RESEARCH EXPERIENCE	University Paris Cité (France) - Visiting researcher. - Research Group MAP5. Applied Mathematics, Probability and Statistics.	5/2024–5/2025
	The Wharton School, University of Pennsylvania (United States) - Postdoctoral Research Fellow. Department of Statistics and Data Science. - Advisors: Pr. Dylan Small and Pr. Sean Hennessy.	11/2020 – 8/2023
	University of Amsterdam (The Netherlands) - PhD Secondment. Department of Clinical Epidemiology and Biostatistics. - Mentor: Pr. Koos Zwinderman.	10/2018 – 12/2018
	University of Bristol (United Kingdom) - PhD Secondment. Department of Population Health Science. - Mentor: Pr. Julian Higgins.	7/2018 – 9/2018
	France Cochrane Center, Université Paris Cité (France) - PhD Secondment. - Mentor: Pr. Raphael Porcher.	7/2017 – 1/2018
	INSERM U1153, Team METHODS, Université Paris Cité (France) - Research Intern. - Mentor: Pr. Raphael Porcher and Dr. Alexandre Vivot.	2/2016 – 7/2016
RESEACRCH FUNDING	French National Research Agency (ANR) - Grant number: 23R09551S-MEDIATION (200k EUR) - Role: Principle investigator. - Topic: Leveraging the use of causal inference methods in evaluating the effectiveness and safety of systemic treatments for psoriasis. - Grant number: ANR-24-CE36-4302 (280k EUR). - Role: Work Package Leader.	

Topic: Shortening the time to confirm or to rebut adverse events of interest related to innovative therapies for immune-mediated inflammatory diseases.

Novo Nordisk (Denmark)

- Grant number: 25R09121C-DEVSTAR (190k EUR).
Role: Principle investigator.
Topic: Causally Interpretable Network Meta-Analysis (CI-NMA): fit-for-purpose methods for decision-making.

HONORS AND AWARDS

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|---|-----------|
| 1. New Faculty Travel Grant Award, Institute of Mathematical Statistics | 8/2022 |
| 2. Junior Researcher Travel Grant, National Science Foundation | 5/2022 |
| 3. Third Prize, 2019 TEDxGhent Research Rally | 11/2019 |
| 4. Norman Breslow Award, American Statistical Association | 8/2019 |
| 5. Student Conference Award, International Society of Clinical Biostatistics | 8/2018 |
| 6. Marie Skłodowska-Curie Actions funding (35.000 EUR/year), project MiRoR | 2016–2019 |
| 7. Ile-de-France International Master Scholarship (10.400 EUR/year) | 2014–2016 |
| 8. French Ministry of Europe and Foreign Affairs Master Scholarship (11.000 E/year) | 2015–2016 |
| 9. Agence Universitaire de la Francophonie Master Scholarship (7.000 EUR/year) | 2014–2015 |
| 10. Valedictorian, Hanoi University of Pharmacy | 6/2014 |
| 11. Full Academic Scholarships, Hanoi University of Pharmacy | 2009–2014 |
| 12. CJ Pharmaceutical Coporation Scholarship | 2013–2014 |

PUBLICATIONS **Statistical Methods**

(*): EQUAL CONTRIBUTION

(†): SENIOR AUTHOR

1. Le K, Sbidian E, **Vo T**[†]. Instrumented difference-in-differences with case-control sampling. Under review.
2. Beclin MF, Le K, **Vo T**[†]. Causal meta-mediation analysis with missing-at-random mediator and outcome data. Under review.
3. Beclin MF, Le K, **Vo T**[†]. Federated estimation of total and indirect treatment effects for survival outcomes. Under review.
4. **Vo T**, Le K, Afach S, Vansteelandt S. Integration of aggregated data in causally interpretable meta-analysis by inverse weighting. *Biometrics*. In press.
5. Le K, Beclin MF, Afach S, **Vo T**[†]. Transportability of aggregate trial results to an external environment in causally interpretable meta-analysis. *Journal of the Royal Statistical Society, Series A*. In press.
6. Gilbert B, Diaz I, Rudolph K, **Vo T**[†]. A novel decomposition to explain heterogeneity in observational and randomized studies of causality. *Biostatistics*. In press.
7. **Vo T**, Williams NT, Liu R, Rudolph KE, Diaz I. Recanting twins: addressing intermediate confounding in mediation analysis. *Stat Med*. 2026;45(3). doi:10.1002/sim.70432.
8. **Vo T**, Ye T, Ertefaie A, Roy S, Flory J, Hennessy S, Vansteelandt S, Small D. Marginal Structural Models for Instrumented Difference in Difference design. *Electron J Statist*. 2024;18(2):5132-5155. doi: 10.1214/24-EJS2313.
9. Roy S, Ye T, Ertefaie A, **Vo T**, Flory J, Hennessy S, Small D. Group Sequential Testing under Instrumented Difference-in-Difference approach. *Stat Med*. 2023;42(21):3838-3859. doi:10.1002/sim.9836.
10. **Vo T**, Davies H, Hackett R, Vansteelandt S. Longitudinal mediation analysis of time-to-event outcome in the presence of competing risk. *Lifetime Data Anal*. 2022;28:380–400. doi:10.1007/s10985-022-09555-7.
11. **Vo T**, Porcher R, Vansteelandt S. Assessing the impact of case-mix heterogeneity in individual participant data meta-analysis: Novel use of I2 statistic and prediction interval. *Research Methods in Medicine & Health Sciences*. 2021;2(1):12-30.doi:10.1177/2632084320957207.
12. **Vo T**, Porcher R, Chaimani A, Vansteelandt S. A Novel Approach for Identifying and Addressing Case-Mix Heterogeneity in Individual Participant Data Meta-Analysis. *Res Synth Methods*. 2019;10(4):582-596. doi:10.1002/jrsm.1382.

Epidemiological Applications

1. [Vo T], Cashin A, Vanderweele T, MacKinnon D, Preacher K, Rudolph KE, Boutron I, Loh WW, Lee H, Vansteelandt S. Assessing bias in mediation analysis. Under revision.
2. Cashin A*, [Vo T*]. Indirect effects in mediation analyses should be tested for statistical significance. *J Clin Epidemiol.* 2024;172. doi:10.1016/j.jclinepi.2024.111395.
3. Truong B, Tran LAT, Pham T, Nguyen TA, [Vo T[†]]. Population adjustment for indirect treatment comparison in health technology assessment: a methodological systematic review. *Res Syn Meth.* 2023;14(5):660-670. doi:10.1002/jrsm.1653.
4. Nguyen V, Sharp M, Superchi C, Baron G, Glonti K, Blanco D, Olsen M, [Vo T], Olarte C, Neveol A, Hren D, Ravaud P, Boutron I. Environmental influences on biomedical doctoral students' research practices when facing dilemmas: vignette-based randomized control trials. *Sci Rep* 13, 16371 (2023). doi:10.1038/s41598-023-42121-1.
5. [Vo T]. A cautionary note on the use of G-computation in population adjustment. *Res Syn Meth.* 2023;1-4. doi:10.1002/jrsm.1621.
6. van Lancker K*, [Vo T*], Akacha M*. Estimands in health technology assessment: a causal inference perspective. *Stat Med.* 2022;41(28):5577-5585. doi:10.1002/sim.9539.
7. Vuong ML, Tu PHT, Duong KL, [Vo T[†]]. Development of minimum reporting sets of patient characteristics in epidemiological research: a methodological systematic review. *Research Methods in Medicine & Health Sciences.* 2023;1-14. doi:10.1177/26320843231191777.
8. [Vo T], Cashin A, Superchi C, Tu PHT, Nguyen TB, Boutron I, MacKinnon D, Vanderweele T, Lee H, Vansteelandt S. Current quality assessment practice in systematic reviews of mediation studies: an overview of systematic reviews. *J Clin Epidemiol.* 2022;143:37-148. doi:10.1016/j.jclinepi.2021.12.013.
9. [Vo T], Vansteelandt S. Challenges in systematic reviews and meta-analyses of mediation studies. *Am J Epidemiol.* 2022;191(6), 1098-1106. doi:10.1093/aje/kwac028.
10. [Vo T], Superchi C, Boutron I, Vansteelandt S. The conduct and reporting of mediation analysis in recently published randomized controlled trials: results from a methodological systematic review. *J Clin Epidemiol.* 2020;117:78-88. doi:10.1016/j.jclinepi.2019.10.001.
11. [Vo T], Vivot A, Porcher R. Impact of Biomarker-based Design Strategies on the Risk of False-Positive Findings in Targeted Therapy Evaluation. *Clin Cancer Res.* 2018;24(24):6257-6264. doi:10.1158/1078-0432.CCR-18-0328.

Clinical Applications

1. Nguyen TPP, Hennessy S, Brensinger CM, Bilker WB, Dember LM, Miano TA, [Vo T], Willis AW, Leonard CE. Emulated target trials assessing the effect of empiric potassium supplementation on mortality, sudden cardiac arrest and stroke among furosemide initiators. *Br J Clin Pharmacol.* In press.
2. Kpenou FC, Han J, Fadel M, [Vo T], Ju HJ, Lee S, Griffiths CEM, Lim HW, French LE, Flohr C, Parisi R, Jemec GBE, Bae JM, Tran VT, Descatha A, Ezzedine K. Work impact of immune-mediated inflammatory skin diseases: a systematic review and meta-analysis. *European Academy of Dermatology and Venereology.* In press.
3. Le K, [Vo T*,[†]], Sbidian E*,[†]. Serious and out-of-hospital infection risk among psoriasis biologic-new users: a nationwide cohort study. *J Am Acad Dermatol.* In press. doi: 10.1016/j.jaad.2026.01.024.
4. Le K, [Vo T*,[†]], Sbidian E*,[†]. Impact of concomitant methotrexate on the persistence of TNF inhibitors in psoriasis. Under review.
5. Tankovic K, Claudepierre P, [Vo T], Le K, Iggui S, Penso L, Sbidian E, Pina-Vegas L. Assessment of cancer risk according to duration of exposure to targeted therapies in patients with spondyloarthritis: a nationwide cohort study. Under review.
6. [Vo T], Roy S, Ye T, Erterfaie A, Nguyen TPP, Flory J, Leonard CE, Small DS, Hennessy S. Effect of exogenous testosterone on cardiovascular, cerebrovascular, and thromboembolic adverse events: Results of three complementary research designs. *Am J Epidemiol.* doi: 10.1093/aje/kwaf098.
7. Murillo C, Cerezo-Téllez E, Torres-Lacomba M, Pham TQ, Lluch E, Falla D, [Vo T[†]]. Unraveling the mechanisms behind the short-term effects of dry needling: new insights from a mediation analysis with repeatedly measured mediators and outcomes. *Arch Phys Med Rehabil.* 2024;105(12):2269-2276. doi: 10.1016/j.apmr.2024.07.016.

8. Hackett R, **Vo T**, Vansteelandt S, Davies-Kershaw H. The role of loneliness on hearing ability and dementia: a novel mediation approach. *J Am Geriatr Soc.* 2023;71(9):2834-2844. doi:10.1111/jgs.183962844.
9. Murillo C, **Vo T**, Vansteelandt S, Harrison L, Cagnie B, Coppieters I, Chys M, Timmers I, Meeus M. How and for whom do cognitive behavioral therapies for chronic pain work? A systematic review and meta-analysis of specific moderators and mediators of treatment outcome in musculoskeletal pain. *Clin Psychol Rev.* 2022;94:102160. doi:10.1016/j.cpr.2022.102160.

ONGOING PROJECT

Statistical Methods

1. **Vo T**, Chambaz A. Causal optimal transport of treatment effect to a target population with limited individual-level data.
2. Roma E, Beclin MF, Remiro-Azocar A, Shu Y, **Vo T**[†]. Improving estimation efficiency for matching-adjusted indirect comparisons.
3. **Vo T**, Nguyen L, Le K, Yu R, Small D. Counter-matching and matching in nested case-control sampling.
4. Beclin MF, Diaz I, **Vo T**[†]. Targeted difference-in-difference learning for survival outcomes.
5. Beclin MF, **Vo T**[†]. Federated causally interpretable meta-analysis.

Epidemiological and Clinical Applications

1. Tu PHT, Le K, Ngo L, **Vo T**[†]. On the magnitude, sign and ranking of recanting-twin path-specific effects.
2. Mauffette N, **Vo T**, Lefebvre G. Bias of non-causal approaches in causal mediation analysis with non-linearities.

BOOK CHAPTERS

1. Nguyen TL, Rohmann JL, **Vo T**, Platt RW, Debray TP. Understanding and defining causal effects. In *Comparative Effectiveness and Personalized Medicine Research Using Real-World Data*. Edited by Debray T, Nguyen TL, Platt R. Chapman & Hall/CRC Handbooks of Modern Statistical Methods.

PROFESSIONAL SERVICES

Editorial activities & professional memberships

1. International Journal of Biostatistics. Associate Editor. 3/2023-now
2. European Federation of Pharmaceutical Industries and Associations & European Federation of Statisticians in the Pharmaceutical Industry Working Group on Estimands for Meta-Analysis. 10/2025-now

Peer-review activities

Journal of the American Statistical Association (×2); Biometrics (×6); Biometrika, Journal of Causal Inference; Journal of the Royal Statistical Society Series C, International Journal of Biostatistics; Research Synthesis Methods (×2); Epidemiology (×2); Clinical Trials; BMC Biomedical Research Methodology (×2); Biometrical Journal, Value in Health, Aging and Mental Health, Multivariate Behavioral Research, Statistics in Medicine, Journal of Computational and Graphical Statistics.

TEACHING EXPERIENCE

Semester-long courses

1. **Lecturer in charge**. Causal Inference. Autumn 2024, 2025
MSc. Public Health Research. University Paris Saclay.
2. **Teaching assistant**. Survival Analysis. Spring 2020
MSc Statistical Data Analysis. Ghent University.
3. **Co-lecturer**. Advanced Biostatistics. Autumn 2017
University Paris Cité.

Short courses and workshops

1. A Gentle Introduction To Causal Diagrams. 10/2024
College of Health Sciences. Vin University.
2. Causal Inference in Medical Research. 04/2024
Graduate School of Public Health. University Paris Saclay.

3. Introduction to Clinical Trials and Evidence Synthesis. 7/2016; 7/2017
Summer school of clinical trials. Hanoi University of Pharmacy.

RESEARCH
ADVISING

Postdocs

1. Enrico Roma. Biostatistics. 2025 - 2028
Topic: Causally interpretable network meta-analysis.
Coadvisor: Antonio Remiro Azocar (Novo Nordisk, 10%).
2. Marie-Félicia Beclin. Biostatistics. 2025 - 2027
Topic: Causal meta-mediation analysis and federated learning.

Doctoral students

1. Khoi Le. Biostatistics. 2023–2026
Topic: Efficacy and safety of biologic treatments in psoriasis: causal inference from observational claim data.
Coadvisor: Emilie Sbidian (University Paris Est Créteil, 50%).

Master students and mentees

1. Aganze Baleke Jonathan (student). MSc. Applied Economics. 2025-2026
University Paris-Est Créteil. Coadvisor: Marie-Felicia Beclin (50%).
Topic: Federated causally interpretable meta-analysis.
2. Lam Nguyen (student). MSc. Public Health. 2025-2026
University Paris Saclay. Coadvisor: Khoi Le (50%).
Topic: Addressing confounding in counter-matched nested case-control design.
3. Long Ngo (student). MSc. Quantitative Epidemiology. 2025–2026
Hasselt University. Topic: On the sign and ranking of recanting-twin path-specific effects.
4. Nicolas Mauffette (student). MSc. Biostatistics. 2024–now
University of Québec at Montréal. Coadvisor: Geneviève Lefebvre (50%).
Topic: Bias of non-causal approaches in mediation analysis under non-linearities.
5. Bang Truong (mentee). PhD. Epidemiology. 2021–2022
Auburn University. Topic: A systematic review of population adusted-indirect comparisons in health technology assessment.
First placement: Real-world evidence data scientist, Abbvie, USA.
6. Luong Vuong (mentee). MSc. Epidemiology. 2021–2022
Antwerp University. Topic: Methods for developing minimum reporting sets of patient characteristics in epidemiological research.
First placement: PhD. candidate in epidemiology, KU Leuven, Belgium.

PhD dissertation committee

1. Examiner for Arnaud Serret-Larmande. Biostatistics. 12/2025
Sorbonnes University. Topic: Advancing population-adjusted indirect comparisons: methods, assumptions and applications.

CONFERENCE
& SEMINAR

Invited talks

1. Asia Pacific Causal Inference Conference. Tianjin, China. 7/2026
Causal optimal transport of treatment effect to a target population with limited individual-level data.
2. 33rd International Biometric Conference. Seoul, Korea. 7/2026
Causally interpretable meta-analysis of mediation analyses.
3. The European Conference on AI for Clinical Applications. Brussels, Belgium. 3/2026
Causal transport of treatment effects to target populations for health technology assessment.
4. 19th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics). London, UK. 12/2025
Targeted learning of treatment effect across populations by optimal transport.
5. 2024 IMS International Conference on Statistics and Data Science, Nice, France. 12/2024
Causally interpretable meta-analysis.

6. Mediation Research Days Conference. Montreal, Canada. 9/2024
Recanting twins: addressing intermediate confounding in mediation analysis.
Chair of the working group "ROBIMA: a quality assessment tool for mediation analysis".
7. Asia-Pacific Causal Inference Conference. Shanghai, China. 7/2024
Recanting twins: addressing intermediate confounding in mediation analysis.
Chair of the online session 1.
8. Joint Congress of Cochrane Skin Group and European Dermato-Epidemiology Network. Paris, France. 4/2024
Population-adjusted indirect comparison in health technology assessment: methods, challenges and current practice.
9. International Society for Health Economics and Outcomes Research (ISPOR) Student Chapter Meeting, Auburn University. 10/2022
Population-adjusted indirect comparisons in health technology assessment

Contributed talks at conferences and departmental seminars

1. European Causal Inference Meeting, Oxford, United Kingdom. 4/2026
Causal optimal transport of treatment effect to a target population with limited individual-level data.
2. PreMeDiCaL Inria-Inserm team, University of Montpellier. 10/2025
Targeted learning of treatment effect across populations by optimal transport.
3. European Causal Inference Meeting, Ghent, Belgium. 4/2025
Causally interpretable meta-analysis under restricted access to individual-level data.
4. Causality and Missing Data Group Meeting, North Carolina State University. 2/2024
Transportability under restricted access to individual-level data: methods and applications in meta-analysis.
5. Epidemiology, Biostatistics and Clinical Trial Unit, Pitié Salpêtrière Hospital. 1/2024
A propensity score weighting approach to integrate aggregated data in random-effect individual-level data meta-analysis.
6. Department of Biostatistics Seminar, Vanderbilt University Medical Center. 2/2023
Causal methods for treatment effect estimation and heterogeneity assessment using real-world data.
7. Division of Pharmacoepidemiology and Pharmacoeconomics Seminar, Brigham's Women Hospital, Harvard Medical School. 2/2023
Causal methods for treatment effect estimation and heterogeneity assessment using real-world data.
8. Real-World Health Navigator (RWHN) Impact Engine Initiative, Northeastern University. 2/2023
Causal methods for treatment effect estimation and heterogeneity assessment using real-world data.
9. American Causal Inference Conference, University of California - Berkeley. 5/2022
Heterogeneity assessment in causal data fusion problems
10. Center of Causal Inference Seminar, University of Pennsylvania. 3/2022
Heterogeneity assessment in causal data fusion problems
11. Center for Research on Healthcare Seminar, University of Pittsburgh. 6/2022
Heterogeneity assessment in causal data fusion problems
12. Division of Biostatistics Seminar, Department of Pharmacology, Physiology and Cancer Biology. Thomas Jefferson University. 8/2022
Heterogeneity assessment in causal data fusion problems
13. IMS New Researchers Conference, George Mason University. 8/2022
Heterogeneity assessment in causal data fusion problems
14. American Causal Inference Conference. University of California - Berkeley. 5/2022
Structural Mean Models for Instrumented Difference-in-Difference Design
15. Annual Conference of the International Society for Clinical Biostatistics. Leuven, Belgium. 7/2019
Novel approaches to address case-mix heterogeneity in meta-analysis

16. Joint Statistical Meeting, American Statistical Association. Denver, Colorado. 7/2019
Novel approaches to address case-mix heterogeneity in meta-analysis
17. Francophone Conference in Clinical Epidemiology. Toulouse, France. 5/2019
Novel approaches to address case-mix heterogeneity in meta-analysis
18. Cochrane Colloquium. Edinburgh, UK. 9/2019
Novel approaches to address case-mix heterogeneity in meta-analysis
19. Joint International Society for Clinical Biostatistics and Australian Statistical Conference. Melbourne, Australia. 8/2018
Rethinking meta-analysis: addressing problems of non-transportability when combining treatment effects across patient populations.
20. Annual Meeting of the Society of Research Synthesis Methodology. Bristol, UK. 7/2018
Rethinking meta-analysis: addressing problems of non-transportability when combining treatment effects across patient populations.
21. European Causal Inference Meeting. Florence, Italy. 4/2018
Rethinking meta-analysis: addressing problems of non-transportability when combining treatment effects across patient populations.